

Aditya ML 2

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## ML LAB 2
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## Aditya Bhende
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## C-C2
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## 28
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```
In [3]
```

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

```
In [4]
```

```
df = pd.read_csv("cardekho.csv")
```

```
In [5]
```

```
df.head()
```

Output:

	name	year	selling_price	km_driven	fuel	\
0	Maruti Swift Dzire VDI	2014	450000	145500	Diesel	
1	Skoda Rapid 1.5 TDI Ambition	2014	370000	120000	Diesel	
2	Honda City 2017-2020 EXi	2006	158000	140000	Petrol	
3	Hyundai i20 Sportz Diesel	2010	225000	127000	Diesel	
4	Maruti Swift VXi BSIII	2007	130000	120000	Petrol	

	seller_type	transmission	owner	mileage(km/ltr/kg)	engine	\
0	Individual	Manual	First Owner	23.40	1248.0	
1	Individual	Manual	Second Owner	21.14	1498.0	
2	Individual	Manual	Third Owner	17.70	1497.0	
3	Individual	Manual	First Owner	23.00	1396.0	
4	Individual	Manual	First Owner	16.10	1298.0	

	max_power	seats
0	74	5.0
1	103.52	5.0
2	78	5.0
3	90	5.0
4	88.2	5.0

```
In [6]
```

```
df.describe()
```

Output:

	year	selling_price	km_driven	mileage(km/ltr/kg)	\
count	8128.000000	8.128000e+03	8.128000e+03	7907.000000	
mean	2013.804011	6.382718e+05	6.981951e+04	19.418783	
std	4.044249	8.062534e+05	5.655055e+04	4.037145	
min	1983.000000	2.999900e+04	1.000000e+00	0.000000	
25%	2011.000000	2.549990e+05	3.500000e+04	16.780000	
50%	2015.000000	4.500000e+05	6.000000e+04	19.300000	
75%	2017.000000	6.750000e+05	9.800000e+04	22.320000	

max	2020.000000	1.000000e+07	2.360457e+06	42.000000
-----	-------------	--------------	--------------	-----------

	engine	seats
count	7907.000000	7907.000000
mean	1458.625016	5.416719
std	503.916303	0.959588
min	624.000000	2.000000
25%	1197.000000	5.000000
50%	1248.000000	5.000000
75%	1582.000000	5.000000
max	3604.000000	14.000000

In [7]

```
df.isnull().sum()
```

Output:

```
name          0
year          0
selling_price  0
km_driven     0
fuel          0
seller_type   0
transmission  0
owner         0
mileage(km/ltr/kg)  221
engine        221
max_power     215
seats         221
dtype: int64
```

In [8]

```
df.shape
```

Output:

```
(8128, 12)
```

In [9]

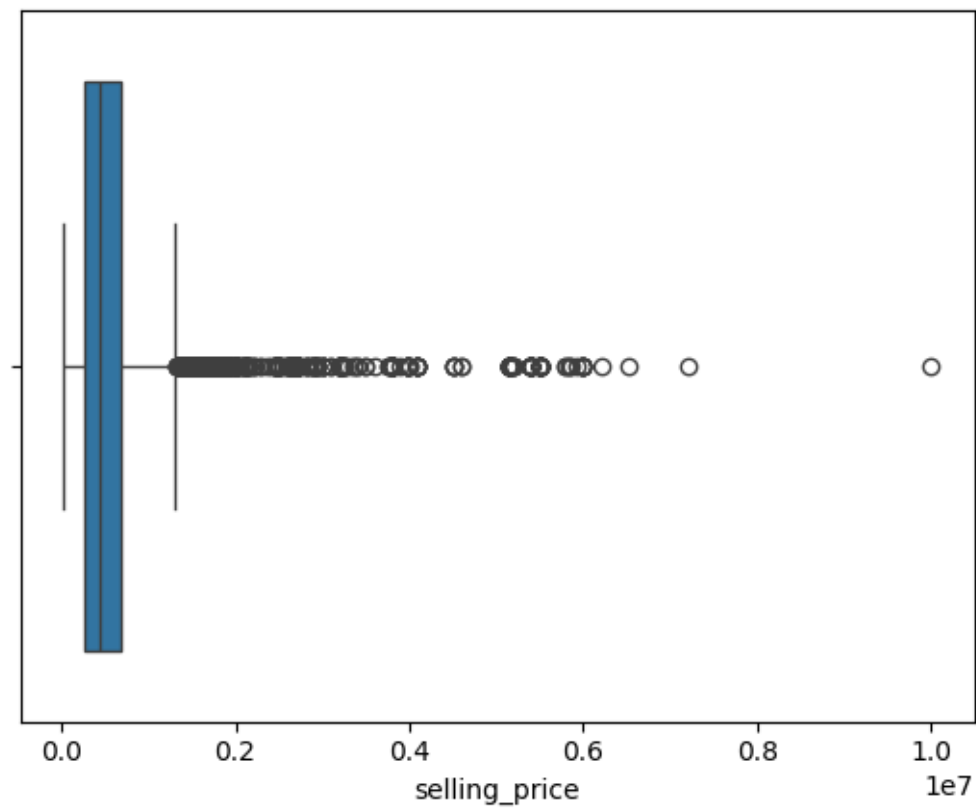
```
df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8128 entries, 0 to 8127
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   name                  8128 non-null   object
1   year                  8128 non-null   int64
2   selling_price         8128 non-null   int64
3   km_driven              8128 non-null   int64
4   fuel                  8128 non-null   object
5   seller_type           8128 non-null   object
6   transmission          8128 non-null   object
7   owner                 8128 non-null   object
8   mileage(km/ltr/kg)    7907 non-null   float64
9   engine                7907 non-null   float64
10  max_power             7913 non-null   object
11  seats                 7907 non-null   float64
dtypes: float64(3), int64(3), object(6)
memory usage: 762.1+ KB
```

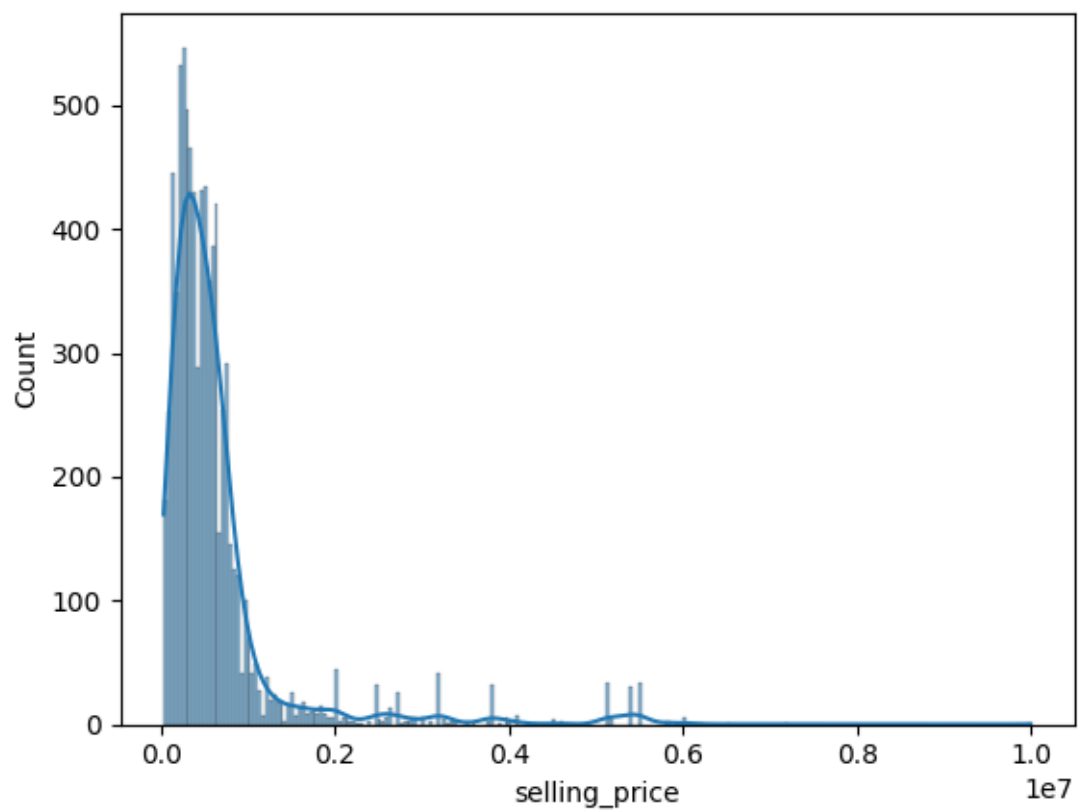
In [10]

```
sns.boxplot(x=df["selling_price"])
plt.show()
```



In [11]

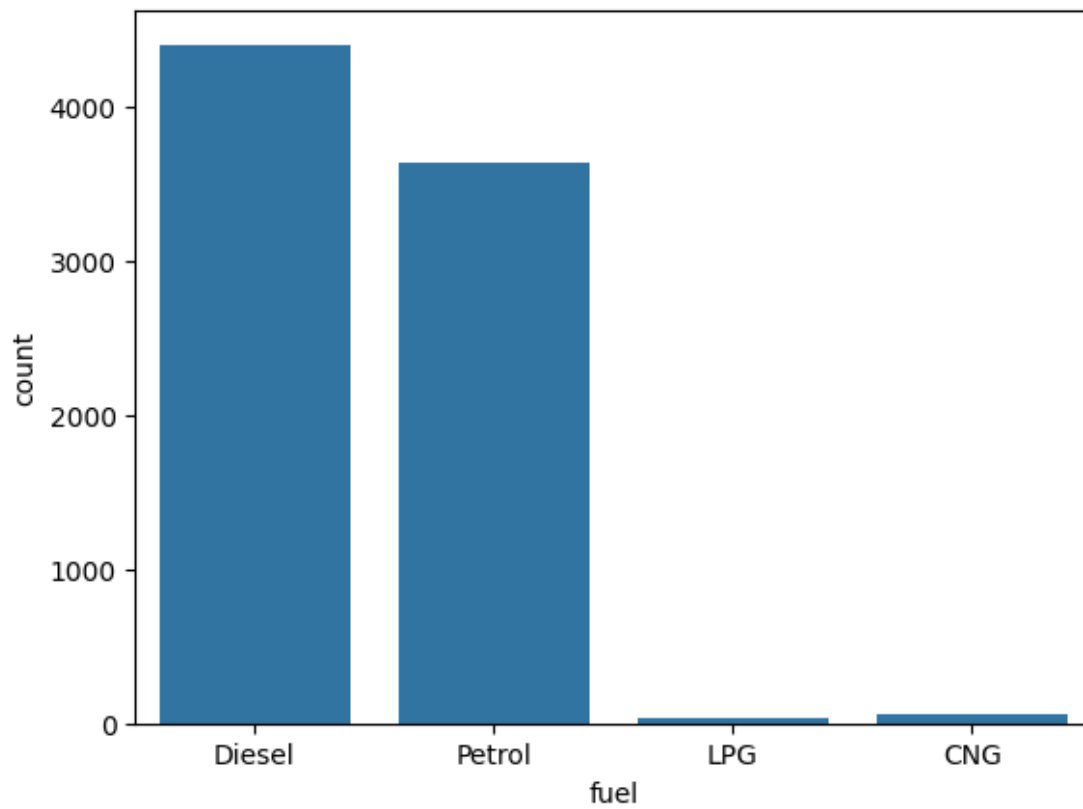
```
sns.histplot(df["selling_price"], kde=True)
plt.show()
```



In [12]

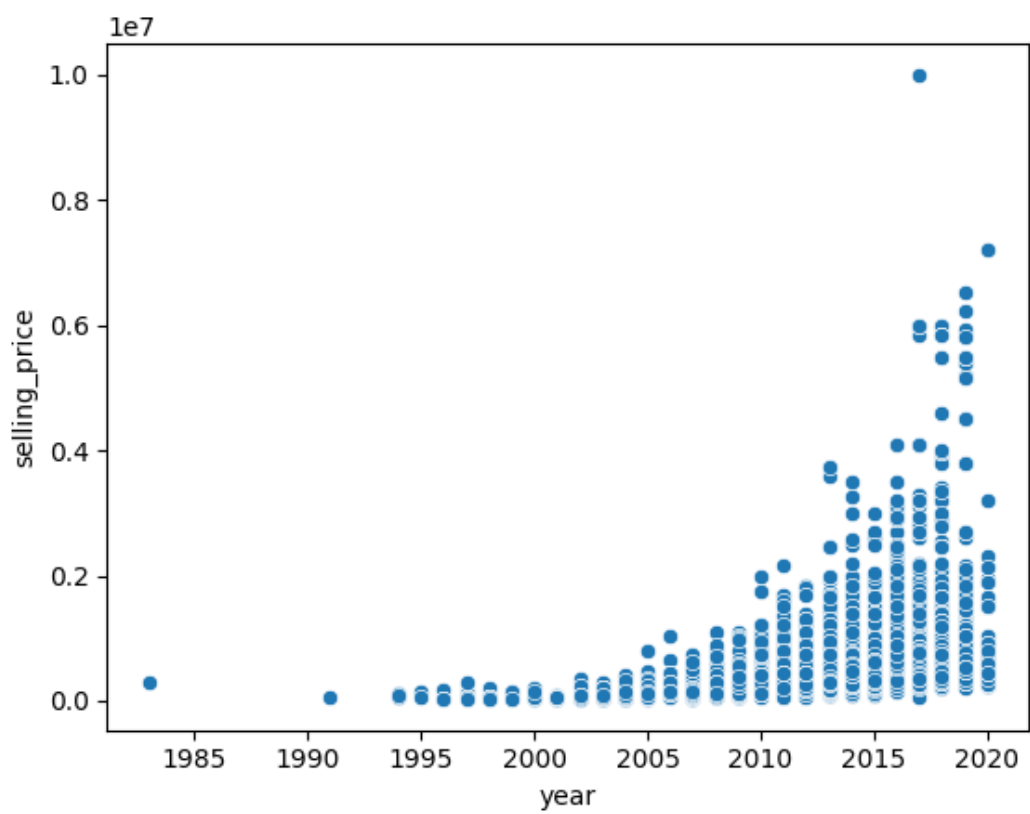
```
sns.countplot(x="fuel", data=df)
```

```
plt.show()
```



In [13]

```
sns.scatterplot(x="year",y="selling_price",data=df)  
plt.show()
```



In [14]

```
sns.countplot(x="owner", data=df)
plt.show()
```

Error output:

```

[1;36m Cell [1;32mIn[70], line 1[1;36m[0m
[1;33m     sns.countplot(x="owner", data=df)[0m
[1;37m                                ^[0m
[1;31mSyntaxError[0m[1;31m:[0m unterminated string literal (detected at line 1)

```

Multiple Linear Regression

In [17]

```

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("cardekho.csv")

df.drop_duplicates(inplace=True)

df.drop("name", axis=1, inplace=True)

df["mileage(km/ltr/kg)"] = df["mileage(km/ltr/kg)"].astype(str).str.replace(" km/kg", "", regex=False)
df["mileage(km/ltr/kg)"] = df["mileage(km/ltr/kg)"].astype(str).str.replace(" kmpl", "", regex=False)
df["mileage(km/ltr/kg)"] = pd.to_numeric(df["mileage(km/ltr/kg)"], errors="coerce")

df["engine"] = df["engine"].astype(str).str.replace(" CC", "", regex=False)
df["engine"] = pd.to_numeric(df["engine"], errors="coerce")

df["max_power"] = df["max_power"].astype(str).str.replace(" bhp", "", regex=False)
df["max_power"] = pd.to_numeric(df["max_power"], errors="coerce")

df.fillna(df.median(numeric_only=True), inplace=True)

le = LabelEncoder()

categorical_cols = ["fuel", "seller_type", "transmission", "owner"]

for col in categorical_cols:
    df[col] = le.fit_transform(df[col].astype(str))

X = df.drop("selling_price", axis=1)
y = df["selling_price"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("MAE :", mean_absolute_error(y_test, y_pred))

```

```
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2 Score:", r2_score(y_test, y_pred))
```

Output:

```
MAE : 173245.83451935017
MSE : 94596219903.37657
RMSE: 307564.98484609165
R2 Score: 0.5686872253789079
```